

Quantum Mechanics in Solids

Unit-5

Lecture-3

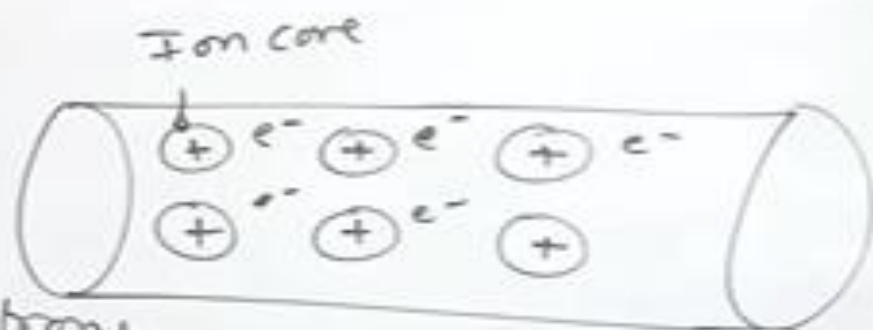
Free electron theory of metals

* Sommerfeld's Theory of Metals

or
Quantum Theory of Metals:

Modification: →

- There is some interaction b/w ion cores & free electrons



$$e^- \quad P.E = 0$$
$$E = k \cdot \epsilon$$

- P.E of free electrons in metals is not zero

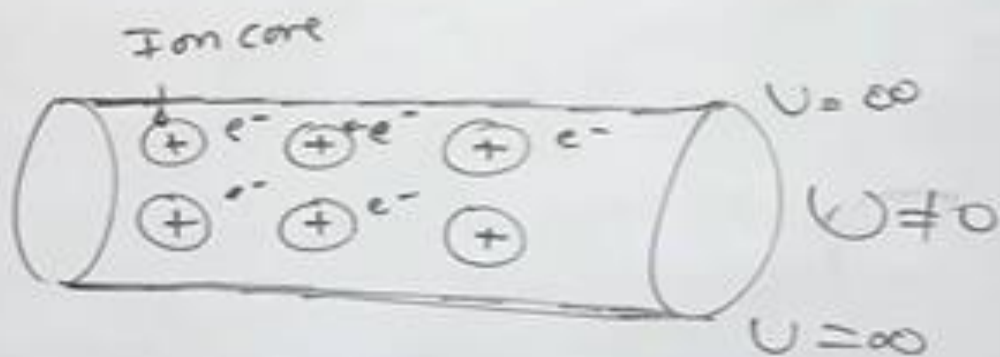
$$(P.E)_{\text{inside}} < (P.E)_{\text{outside}}$$

* Sommerfeld's Theory of Metals

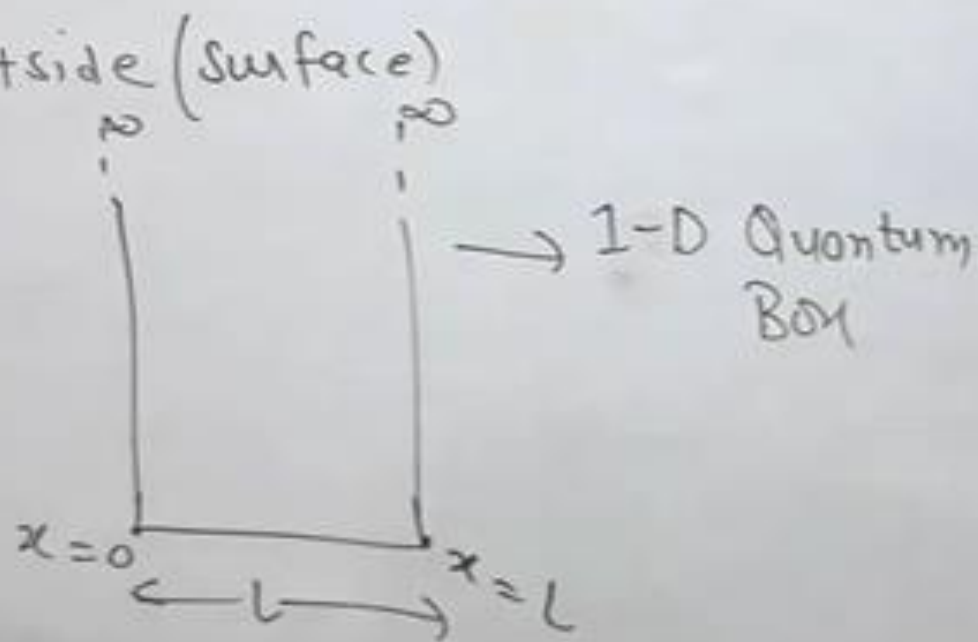
or Quantum Theory of Metals:

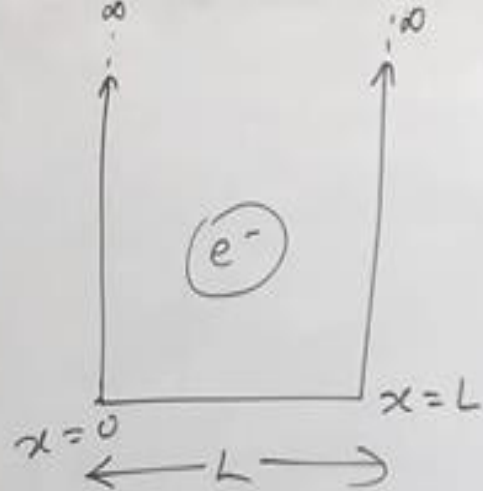
Assumptions:

- P.E of a free electron inside metal is same everywhere



- P.E of free electron outside (surface) the metal is ∞





$$V(x) = 0 \quad 0 \leq x \leq L$$

$$= \infty \quad 0 < x, x > L$$

P.E independent of time

Schrodinger's time independent Eqⁿ

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} (E - V) \psi = 0$$

$$V=0, \text{ for } 0 \leq x \leq L$$

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} E \psi = 0 \quad (1)$$

$$\frac{2mE}{\hbar^2} = K^2$$

$$\frac{d^2\psi}{dx^2} + K^2 \psi = 0 \quad (2)$$

General solⁿ

$$\psi = A \sin Kx + B \cos Kx \quad (3)$$

At

$$x=0, \psi=0$$

$$B=0$$

$$\text{At } x=L, \psi=0$$

$$0 = A \sin KL$$

$$KL = n\pi, n=1,2,3 \dots$$

$$K = \frac{n\pi}{L}$$

$$\psi_n = A \sin\left(\frac{n\pi}{L}x\right) \quad (4)$$

Normalisation condition

$$\int_0^L |\psi(x)|^2 dx = 1$$

$$\int_0^L A^2 \sin^2\left(\frac{n\pi}{L}x\right) dx = 1$$

$$A^2 \int_0^L \sin^2\left(\frac{n\pi}{L}x\right) dx = 1$$

$$A^2 \cdot x \frac{L}{2} = 1$$

$$A = \sqrt{\frac{2}{L}}$$

$$\boxed{\psi_n(x) = \left(\sqrt{\frac{2}{L}}\right) \sin\left(\frac{n\pi}{L}x\right)} \quad (5)$$

$$\frac{d^2\psi}{dx^2} = -\left(\frac{\sqrt{2}}{L}\right) \frac{n^2\pi^2}{L^2} \sin\left(\frac{n\pi}{L}\right)x$$

$$-\left(\frac{\sqrt{2}}{L}\right) \frac{n^2\pi^2}{L^2} \sin\left(\frac{n\pi}{L}\right)x + \frac{2mE}{\hbar^2} \times \left(\frac{\sqrt{2}}{L}\right) \sin\left(\frac{n\pi}{L}\right)x = 0$$

$$-\left(\frac{\sqrt{2}}{L}\right) \frac{n^2\pi^2}{L^2} \sin\left(\frac{n\pi}{L}\right)x = -\frac{2mE}{\hbar^2} \left(\frac{\sqrt{2}}{L}\right) \sin\left(\frac{n\pi}{L}\right)x$$

$$\frac{n^2\pi^2}{L^2} = \frac{2mE}{\hbar^2}$$

$$E_n = \frac{\hbar^2}{2m} \left(\frac{n\pi}{L}\right)^2$$

Fermi Level:

Top-most filled level at Absolute tempⁿ

Fermi Energy (E_F): → Energy corresponds to Fermi level

$$n_F = \frac{N}{2}$$

↑
No of Fermi level

← No of Electron



$$E_F = \frac{\hbar^2}{2m} \left(\frac{n_F \pi}{L} \right)^2$$

$$= \frac{\hbar^2}{2m} \left(\frac{N \pi}{2L} \right)^2$$

$$= \frac{\hbar^2}{4\pi^2 \times 2m} \times \frac{N^2 \pi^2}{4 \times L^2}$$

$$E_F = \frac{\hbar^2}{2m} \left(\frac{N}{4L} \right)^2$$

$$E_F \propto \left(\frac{N}{L} \right)$$

↑
No of Electrons per unit length

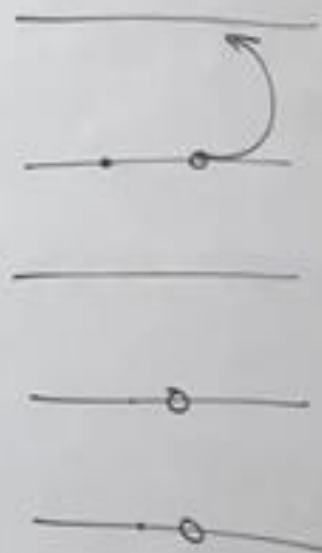
* Density of State for 1-D Box:-

No. of Energy levels
for certain Energy Range.

$$D(E) = \frac{dn}{dE}$$



At E



$E + \Delta E$

* Density of State for 1-D Box:-

No. of Energy levels
for certain Energy
Range

$$D(E) = \frac{dn}{dE}$$

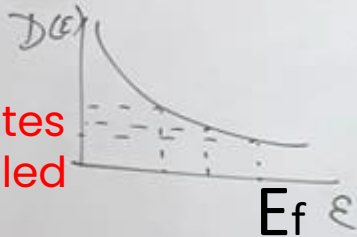
$$E = \frac{\hbar^2}{2m} \left(\frac{n\pi}{L} \right)^2$$

$$E = \frac{\hbar^2}{2m} \left(\frac{\pi}{L} \right)^2 n^2$$

$$E = \frac{\hbar^2}{4\pi^2} \times \frac{1}{2m} \times \frac{\pi^2}{L^2} \times n^2$$

$$E = \frac{\hbar^2}{8mL^2} n^2$$

$$D(E) \propto \frac{1}{\sqrt{E}}$$



It shows that all the states
up to Fermi level are filled
at 0K.

$$\frac{dE}{dn} = \frac{\hbar^2}{4mL^2} \times 2n$$

$$\frac{dE}{dn} = \frac{\hbar^2 n}{4mL^2}$$

$$D(E) = \frac{dn}{dE} = \frac{4mL^2}{n\hbar^2}$$

$$D(E) = \frac{8mL^2}{n\hbar^2}$$

Each k state can hold 2 electrons (of
opposite spins), so the number of
electron states is:

$$D(E) \propto \frac{1}{n}$$



We know that

$$E = \frac{\hbar^2}{2m} \left(\frac{n\pi}{L} \right)^2 = \frac{\hbar^2}{4\pi^2} \times \frac{1}{2m} \times \frac{\pi^2}{L^2} n^2$$

$$E = \frac{\hbar^2 n^2}{8mL^2} \Rightarrow n = \frac{\sqrt{8mL^2 E}}{\hbar}$$

$$D(E) = \frac{8mL^2}{\hbar^2} \times \frac{\hbar}{\sqrt{8mL^2 E}} \Rightarrow$$

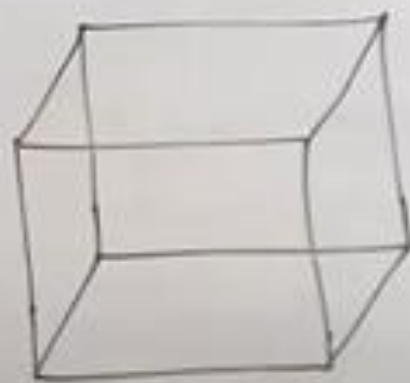
$$D(E) = \frac{\sqrt{8mL^2}}{\hbar \sqrt{E}}$$

Density of State & Energy of Quantum well

* Quantum well: →

one Dimension = Nanoscale

Two Dimension = Normal/Bulk



Bulk



Quantum
well

Density of State & Energy of Quantum well

Density of State for 2 Dimensions

$$\frac{2mE}{\hbar^2} = k^2$$

$$k = \sqrt{\frac{2mE}{\hbar^2}}$$

We know that

Density of State $D(E) = \frac{dN}{dE}$

$N =$ No. of Allowed Energy level

$dE =$ Energy difference

Single-Energy level = Area of First B.Z.

$$= \frac{\pi}{L} \times \frac{\pi}{L} = \frac{\pi^2}{L^2}$$

Allowed Energy level = πk^2

$$N = \frac{\pi k^2}{\frac{\pi^2}{L^2}} \times \frac{1}{2} = \frac{\pi k^2}{\pi^2} \times L^2 \times \frac{1}{2}$$

$$N = \frac{k^2 L^2}{2\pi}$$

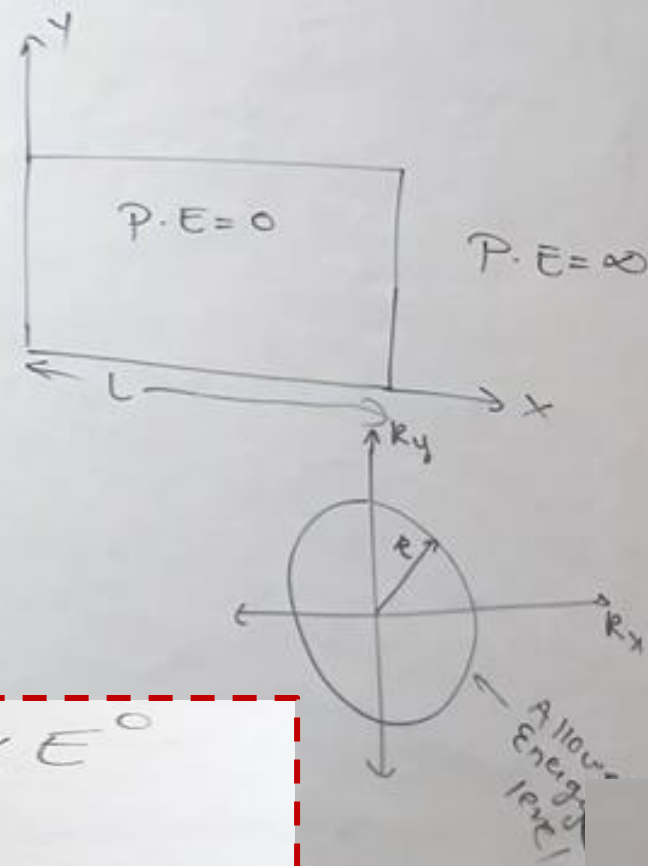
$$N = \frac{2mE}{\hbar^2} \times \frac{L^2}{2\pi}$$

$$N = \frac{mEL^2}{\hbar^2 \pi}$$

$$D(E) = \frac{dN}{dE}$$

$$D(E) = \frac{d}{dE} \left[\frac{mEL^2}{\hbar^2 \pi} \right]$$

$$D(E) = \frac{mL^2}{\pi \hbar^2}$$



$$D(E) \propto E^0$$



Quantum well
or
1-D Nanomaterial.

Fermi Surface & Fermi Energy $\rightarrow$

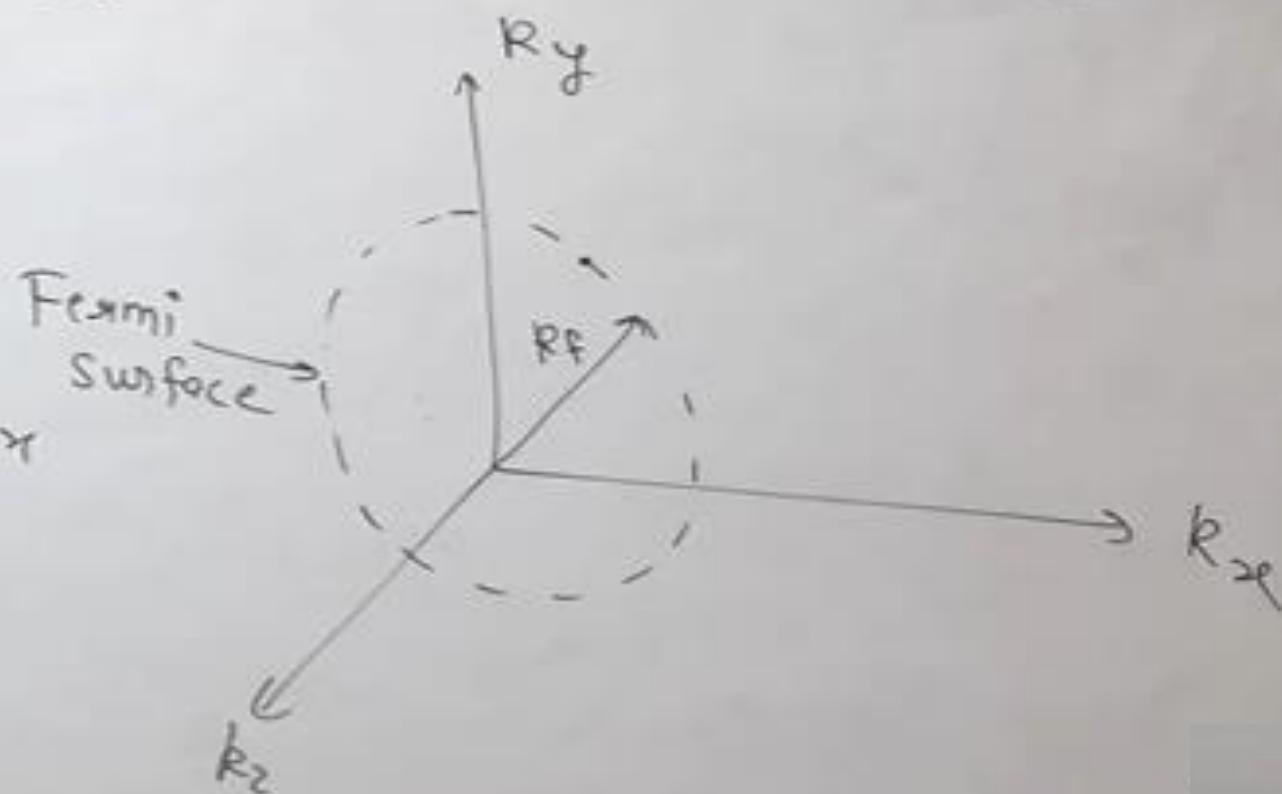
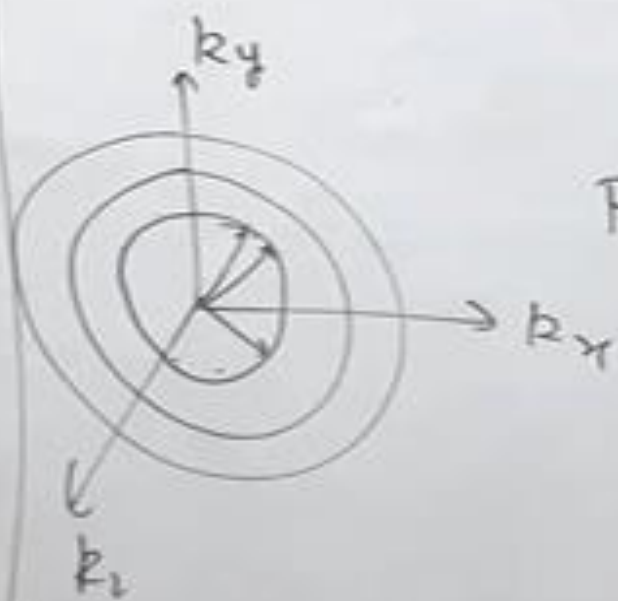
We know that

$$\frac{2m}{\hbar^2} E = k^2$$

$$E = \frac{\hbar^2}{2m} (k_x^2 + k_y^2 + k_z^2) \rightarrow E_{q^n} \text{ Like Sphere}$$

Energy of an electron in energy state

$$x^2 + y^2 + z^2 = r^2$$



Fermi Energy

$$E_F = \frac{\hbar^2}{2m} k_F^2$$

N = no. of Free Electrons in crystal

No. of k -values for N no. of free electrons

$$= \frac{\frac{4}{3} \pi k_F^3}{\left(\frac{2\pi}{L}\right)^3}$$

No. of Energy states for N

No. of free Electrons

$$N = \frac{2 \times \frac{4}{3} \pi k_F^3}{\left(\frac{2\pi}{L}\right)^3}$$

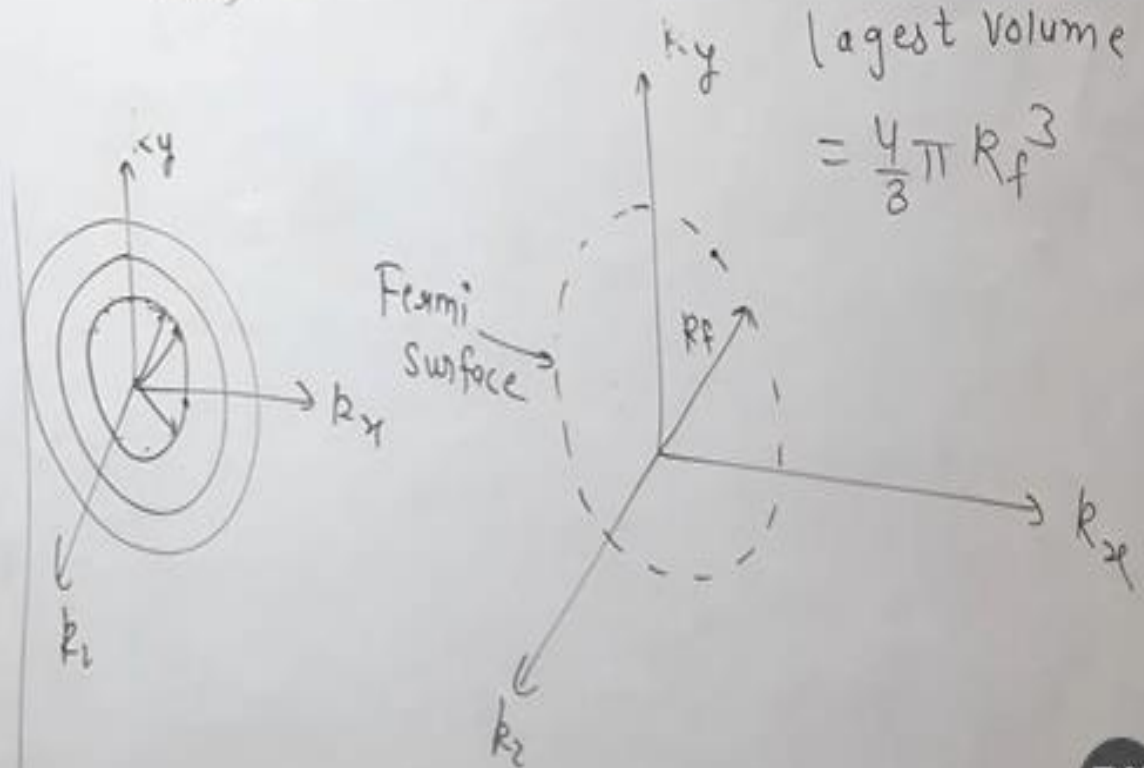
Smallest sphere

$$k_x = \frac{2\pi n}{L} = k_y = k_z$$

$$n=1$$

$$k_x = k_y = k_z = \frac{2\pi}{L}$$

$$\left(\frac{2\pi}{L}\right)^3 = \text{smallest volume}$$



$$N = \frac{2 \times \frac{4}{3} \pi R_f^3}{8 \pi^2} \times \frac{L^3}{2}$$

$$N = \frac{R_f^3}{3 \pi^2} \times L^3$$

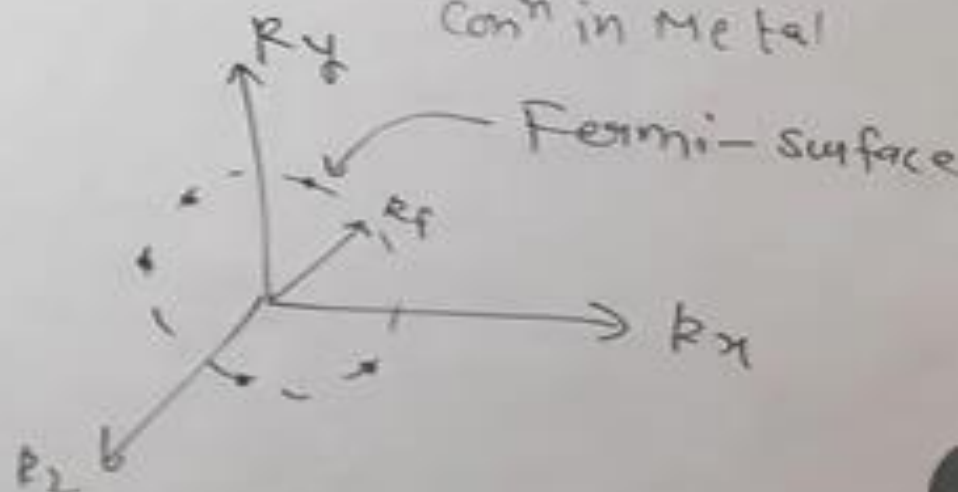
$$R_f^3 = 3 \pi^2 \frac{N}{V}$$

$$R_f = \left(3 \pi^2 \frac{N}{V} \right)^{\frac{1}{3}} \quad \text{--- (4)}$$

$\frac{N}{V}$ = No of Electrons / Unit Volume = Free Electron
Conⁿ in Metal

So, Fermi Energy

$$E_F = \frac{\hbar^2}{2m} \left(3 \pi^2 \frac{N}{V} \right)^{2/3}$$



Density of State: - Density of State for 3 Dimensions

↓
No of filled Energy levels
per unit Energy Range.

$$D(E) = \frac{dN}{dE} = \frac{V}{3\pi^2} \times \left(\frac{2m}{\hbar^2}\right)^{3/2} \times \frac{3}{2} E^{1/2}$$

For 3-D crystal

$$D(E) \propto E^{1/2}$$

$$D(E) = \frac{V}{2\pi^2} \left(\frac{2m}{\hbar^2}\right)^{3/2} E^{1/2}$$

$$D(E) = \frac{dN}{dE}$$

We know that, In General

$$E = \frac{\hbar^2}{2m} \left(\frac{3\pi^2 N}{V}\right)^{2/3}$$

$$N^{2/3} = \left(\frac{V}{3\pi^2}\right)^{2/3} \frac{2m}{\hbar^2} E$$

$$N = \frac{V}{3\pi^2} \times \left(\frac{2m}{\hbar^2}\right)^{3/2} E^{3/2}$$

This formula only applicable $D(0)$
at 0K

At T temp

Number density of filled energy states
at TK

$$N(E) = D(E) \times f(E)$$

↑ Fermi-Dirac Function

$$f(E) = \frac{1}{e^{(E-E_f)/kT} + 1}$$

$$E < E_f, T=0K \quad f(E) = 1$$

$$E > E_f \quad f(E) = 0$$

